

ECO101

Introduction to Microeconomics

Yamim Rashad Nawaz

BRAC University

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Credit: Cengage 2014

Week 5

- What is elasticity? What kinds of issues can elasticity help us understand?
- What is the price elasticity of demand? How is it related to the demand curve? How is it related to revenue & expenditure?
- What is the price elasticity of supply? How is it related to the supply curve?
- What are the income and cross-price elasticities of demand?

A scenario

You design websites for local businesses. You charge \$200 per website, and currently sell 12 websites per month.

Your costs are rising (including the opportunity cost of your time), so you consider raising the price to \$250.

The law of demand says that you won't sell as many websites if you raise your price.

How many fewer websites? How much will your revenue fall, or might it increase?

Elasticity

Basic idea:

Elasticity measures how much one variable responds to changes in another variable.

One type of elasticity measures how much demand for your websites will fall if you raise your price.

Definition:

Elasticity is a numerical measure of the responsiveness of Q^d or Q^s to one of its determinants.

Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

- **Price elasticity of demand** measures how much Q^d responds to a change in P .
- Loosely speaking, it measures the price-sensitivity of buyers' demand.

Price Elasticity of Demand

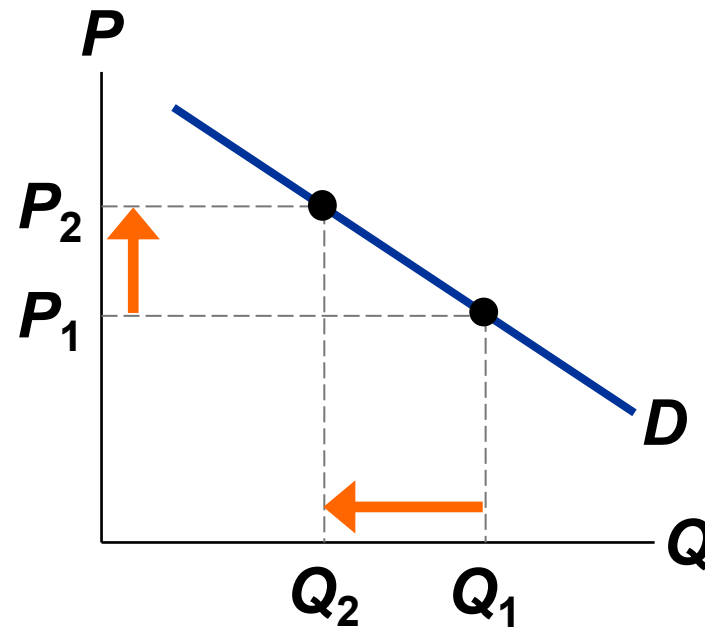
$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

Example:

Price elasticity of demand equals

$$\frac{15\%}{10\%} = 1.5$$

P rises by 10%



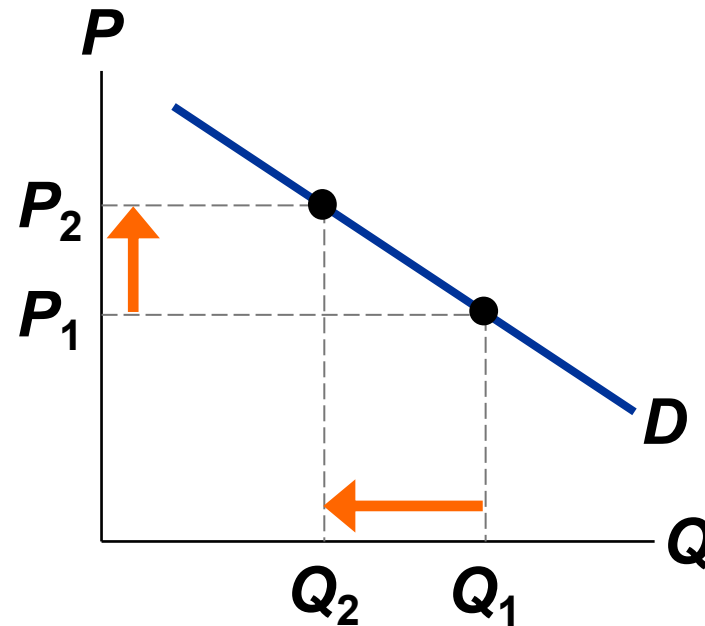
Q falls by 15%

Price Elasticity of Demand

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q^d}{\text{Percentage change in } P}$$

Along a ***D*** curve, ***P*** and ***Q*** move in opposite directions, which would make price elasticity negative.

We will drop the minus sign and report all price elasticities as positive numbers.



Calculating Percentage Changes

- We use the **midpoint method**:

$$\frac{\text{end value} - \text{start value}}{\text{midpoint}} \times 100\%$$

- The midpoint is the number halfway between the start and end values, the average of those values.
- It doesn't matter which value you use as the start and which as the end—you get the same answer either way!

Calculating Percentage Changes

- Using the midpoint method, the % change in **P** equals

$$\frac{\$250 - \$200}{\$225} \times 100\% = 22.2\%$$

- The % change in **Q** equals

$$\frac{12 - 8}{10} \times 100\% = 40.0\%$$

- The price elasticity of demand equals

$$40/22.2 = 1.8$$

Food for thought

Use the following information to
calculate the price elasticity of
demand for hotel rooms:

if $P = \$70$, $Q^d = 5000$

if $P = \$90$, $Q^d = 3000$

Food for thought

Use midpoint method to calculate
% change in Q^d

$$(5000 - 3000)/4000 = 50\%$$

% change in P

$$(\$90 - \$70)/\$80 = 25\%$$

The price elasticity of demand equals

$$\frac{50\%}{25\%} = 2.0$$

What determines price elasticity?

To learn the determinants of price elasticity, we look at a series of examples.

Each compares two common goods.

In each example:

- Suppose the prices of both goods rise by 20%.
- The good for which Q^d falls the most (in percent) has the highest price elasticity of demand. Which good is it? Why?
- What lesson does the example teach us about the determinants of the price elasticity of demand?

EXAMPLE 1:

Breakfast Cereal vs. Sunscreen

The prices of both of these goods rise by 20%.

For which good does Q^d drop the most? Why?

Breakfast cereal has close substitutes (e.g., pancakes, Eggo waffles, leftover pizza), so buyers can easily switch if the price rises.

Sunscreen has no close substitutes, so consumers would probably not buy much less if its price rises.

Lesson: Price elasticity is higher when close substitutes are available.

EXAMPLE 2:

“Blue Jeans”

VS.

“Clothing”

The prices of both goods rise by 20%.

For which good does Q^d drop the most? Why?

For a narrowly defined good such as blue jeans, there are many substitutes (khakis, shorts, Speedos).

There are fewer substitutes available for broadly defined goods. (There aren't too many substitutes for clothing, other than living in a nudist colony.)

Lesson: Price elasticity is higher for narrowly defined goods than broadly defined ones.

EXAMPLE 3: Insulin vs. Cox's Bazar Cruises

The prices of both of these goods rise by 20%.
For which good does Q^d drop the most? Why?

To millions of diabetics, insulin is a necessity.
A rise in its price would cause little or no
decrease in demand.

A cruise is a luxury. If the price rises,
some people will forego it.

***Lesson: Price elasticity is higher for luxuries
than for necessities.***

EXAMPLE 4:
Fuel in the Short
Run vs.
Fuel in the Long
Run

The price of gasoline rises 20%. Does Q^d drop more in the short run or the long run? Why?

There's not much people can do in the short run, other than ride the bus or carpool.

In the long run, people can buy smaller cars or live closer to where they work.

Lesson: Price elasticity is higher in the long run than the short run.

The Determinants of Price Elasticity: A Summary

The price elasticity of demand depends on:

- the extent to which close substitutes are available
- whether the good is a necessity or a luxury
- how broadly or narrowly the good is defined
- the time horizon—elasticity is higher in the long run than the short run

The Variety of Demand Curves

The price elasticity of demand is closely related to the slope of the demand curve.

Rule of thumb:

The flatter the curve, the bigger the elasticity.

The steeper the curve, the smaller the elasticity.

Five different classifications of *D* curves...

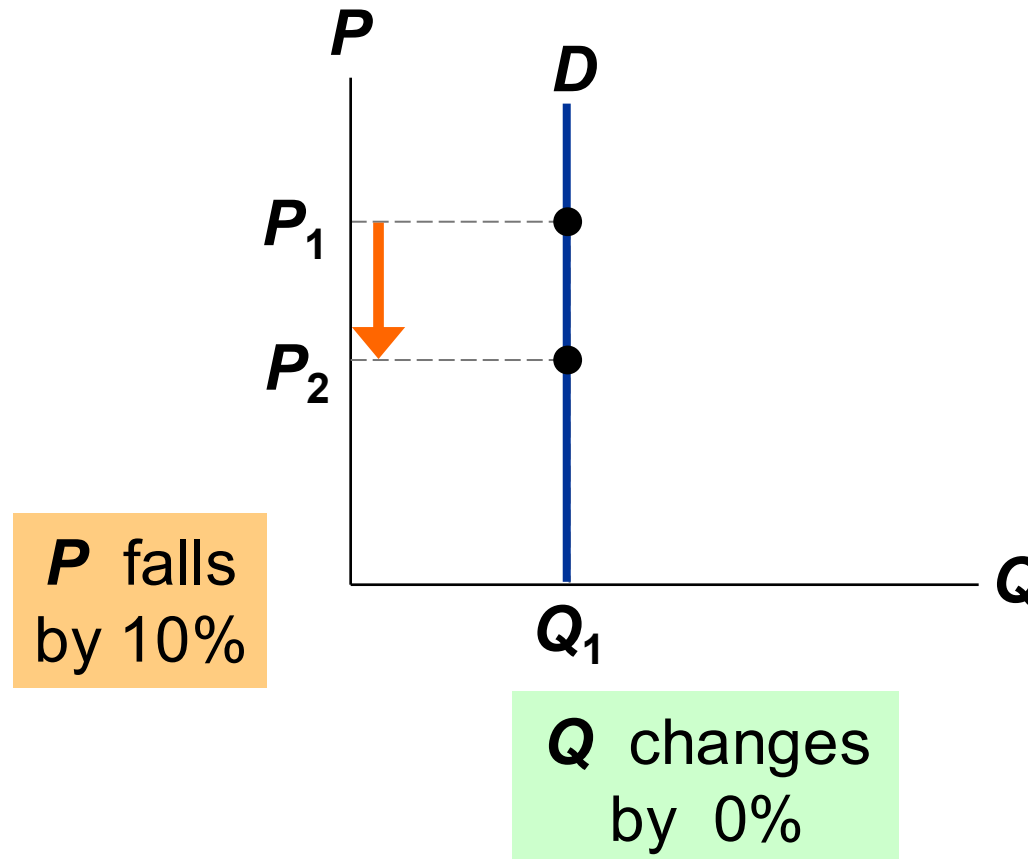
“Perfectly inelastic demand” (one extreme case)

$$\text{Price elasticity of demand} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{0\%}{10\%} = 0$$

D curve:
vertical

Consumers'
price sensitivity:
none

Elasticity:
0



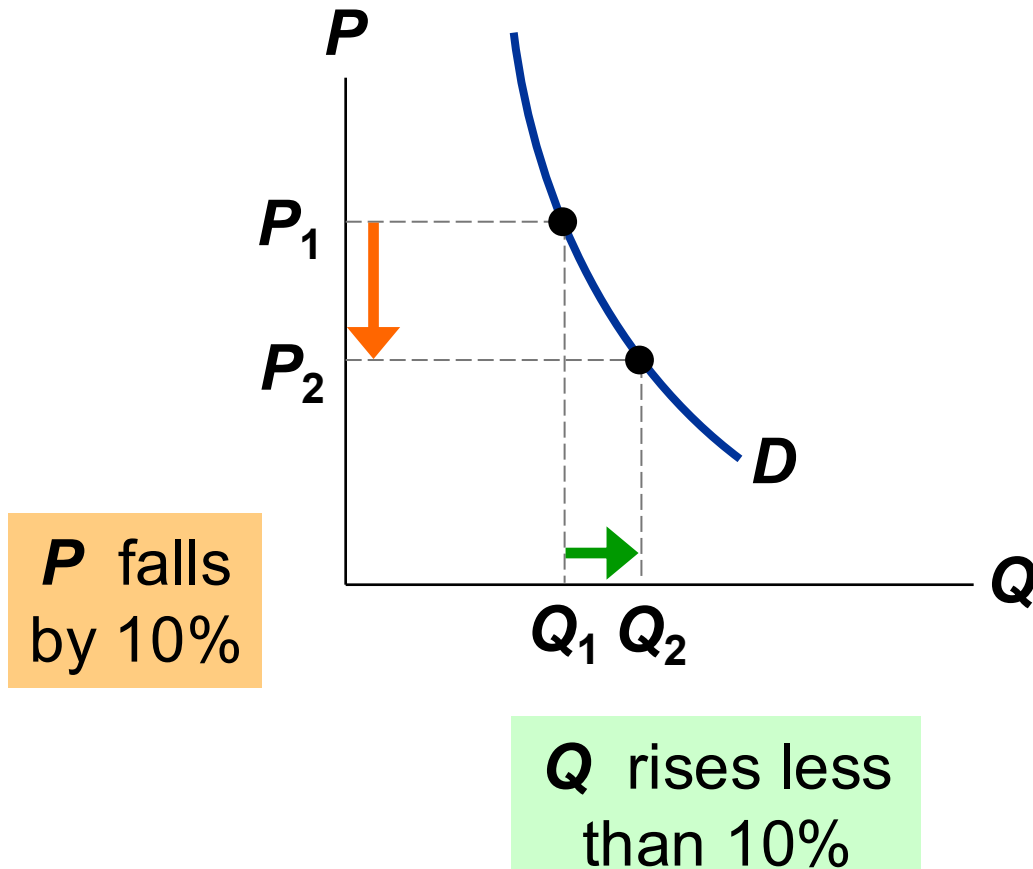
"Inelastic demand"

$$\text{Price elasticity of demand} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{< 10\%}{10\%} < 1$$

D curve:
relatively steep

Consumers'
price sensitivity:
relatively low

Elasticity:
< 1



“Unit elastic demand”

$$\text{Price elasticity of demand} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{10\%}{10\%} = 1$$

D curve:

intermediate slope

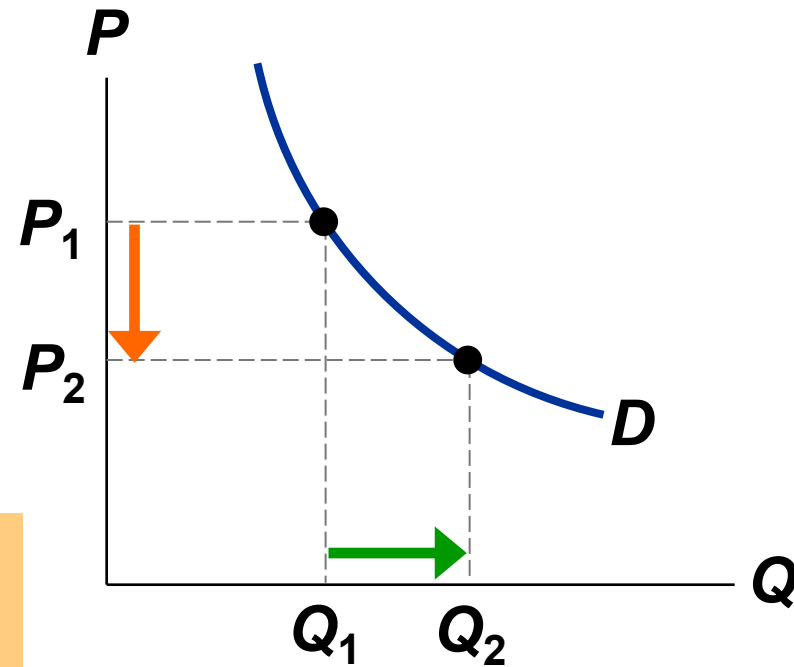
Consumers’
price sensitivity:

intermediate

Elasticity:

1

P falls
by 10%



Q rises by 10%

"Elastic demand"

$$\text{Price elasticity of demand} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{> 10\%}{10\%} > 1$$

D curve:

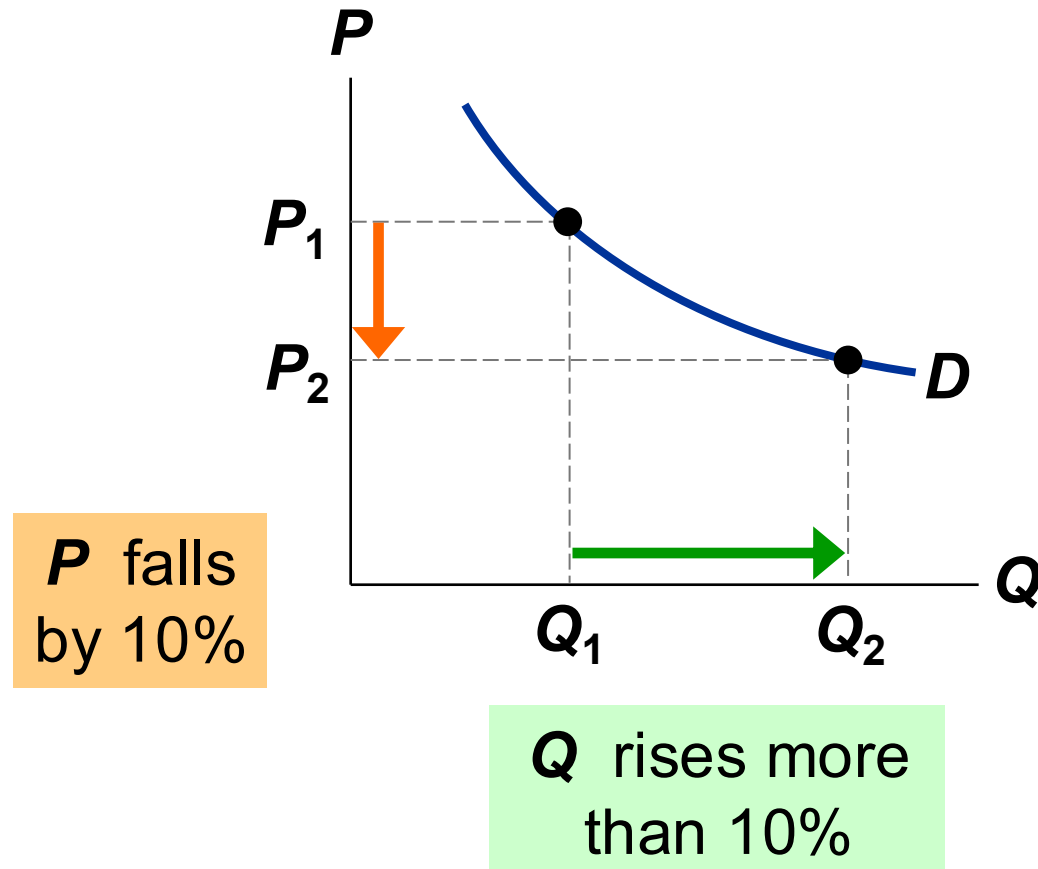
relatively flat

Consumers'
price sensitivity:

relatively high

Elasticity:

> 1



“Perfectly elastic demand” (the other extreme)

$$\text{Price elasticity of demand} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{\text{any } \%}{0\%} = \text{infinity}$$

D curve:

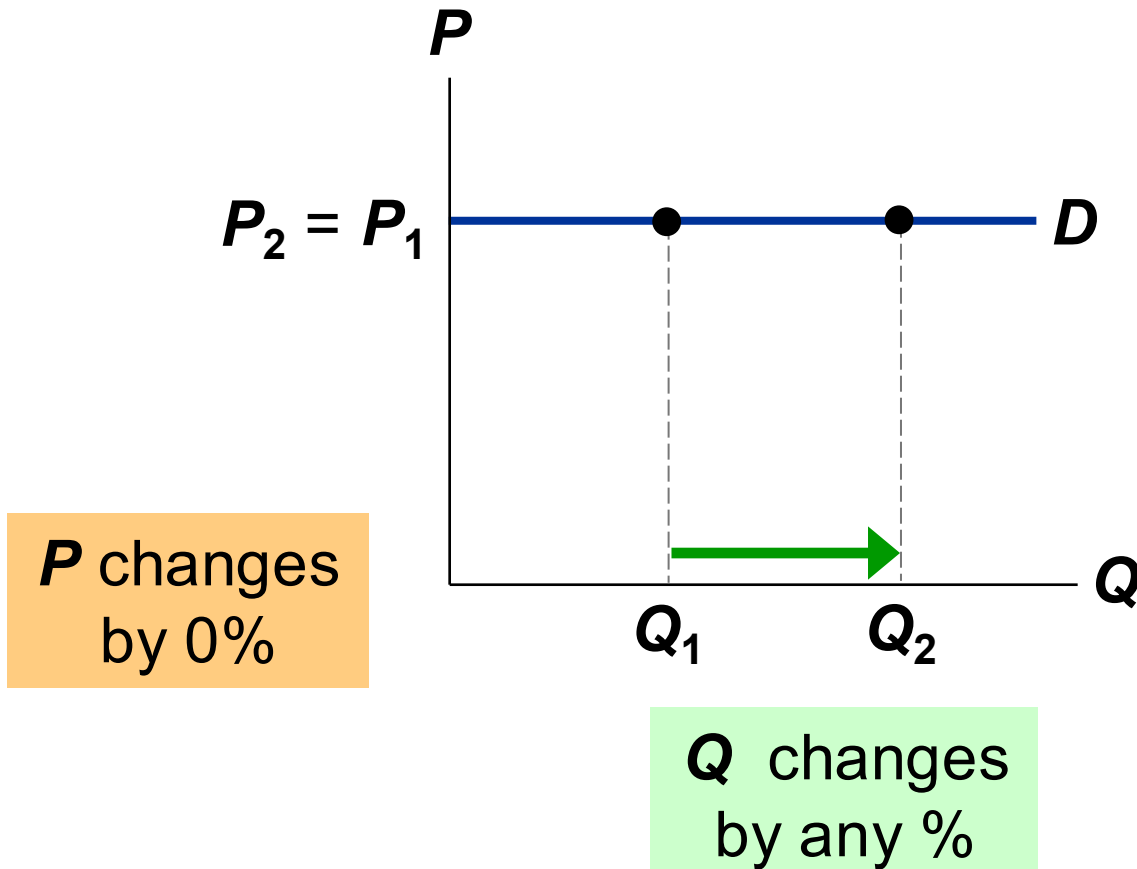
horizontal

Consumers’
price sensitivity:

extreme

Elasticity:

infinity



**A few elasticities
from the real
world**

Eggs	0.1
Healthcare	0.2
Rice	0.5
Housing	0.7
Beef	1.6
Restaurant meals	2.3
Mountain Dew	4.4

Price Elasticity and Total Revenue

Continuing our scenario, if you raise your price from \$200 to \$250, would your revenue rise or fall?

$$\text{Revenue} = P \times Q$$

A price increase has two effects on revenue:

Higher P means more revenue on each unit you sell.

But you sell fewer units (lower Q), due to law of demand.

Which of these two effects is bigger?

It depends on the price elasticity of demand.

Price Elasticity and Total Revenue

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q}{\text{Percentage change in } P}$$

$$\text{Revenue} = P \times Q$$

- If demand is elastic, then
price elast. of demand > 1
 $\% \text{ change in } Q > \% \text{ change in } P$
- The fall in revenue from lower Q is greater than the increase in revenue from higher P , so revenue falls.

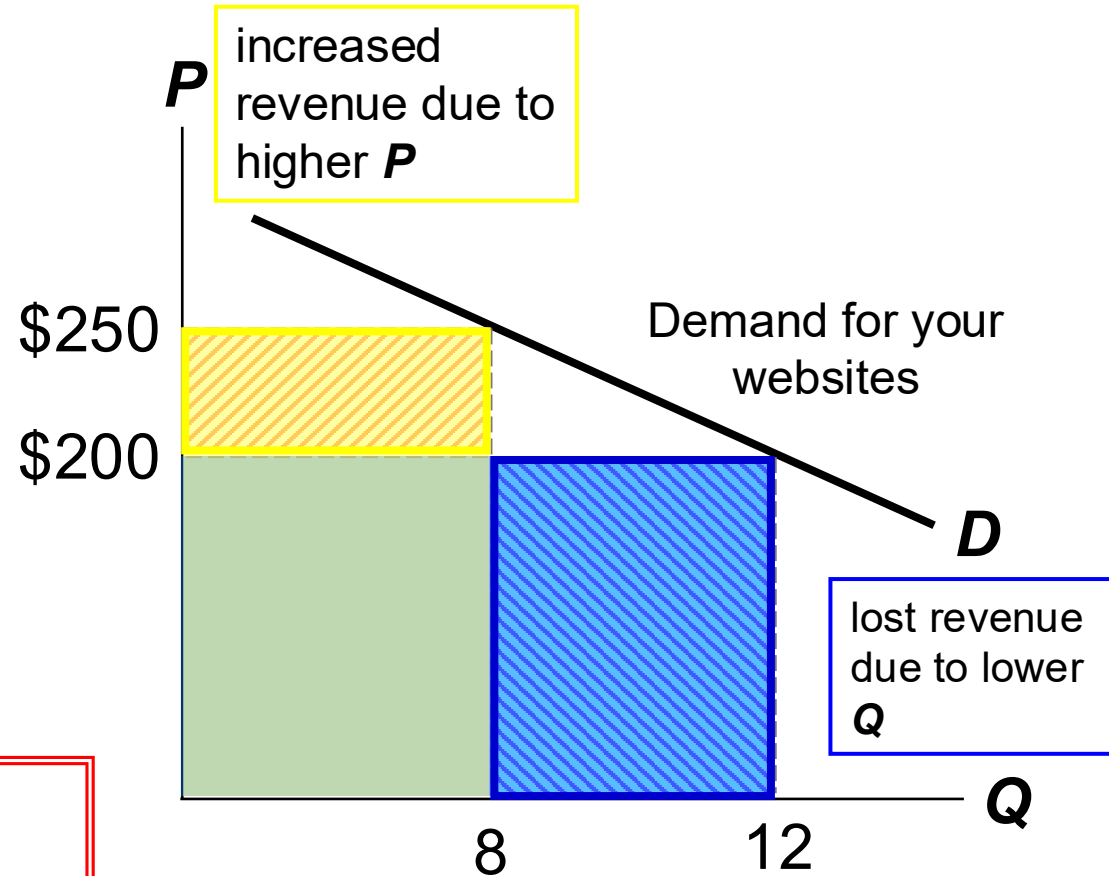
Price Elasticity and Total Revenue

Elastic demand
(elasticity = 1.8)

If $P = \$200$,
 $Q = 12$ and
revenue = \$2400.

If $P = \$250$,
 $Q = 8$ and
revenue = \$2000.

When D is elastic,
a price increase
causes revenue to fall.



Price Elasticity and Total Revenue

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in } Q}{\text{Percentage change in } P}$$

$$\text{Revenue} = P \times Q$$

- If demand is inelastic, then
price elast. of demand < 1
 $\% \text{ change in } Q < \% \text{ change in } P$
- The fall in revenue from lower Q is smaller than the increase in revenue from higher P , so revenue rises.
- In our example, suppose that Q only falls to 10 (instead of 8) when you raise your price to \$250.

Price Elasticity and Total Revenue

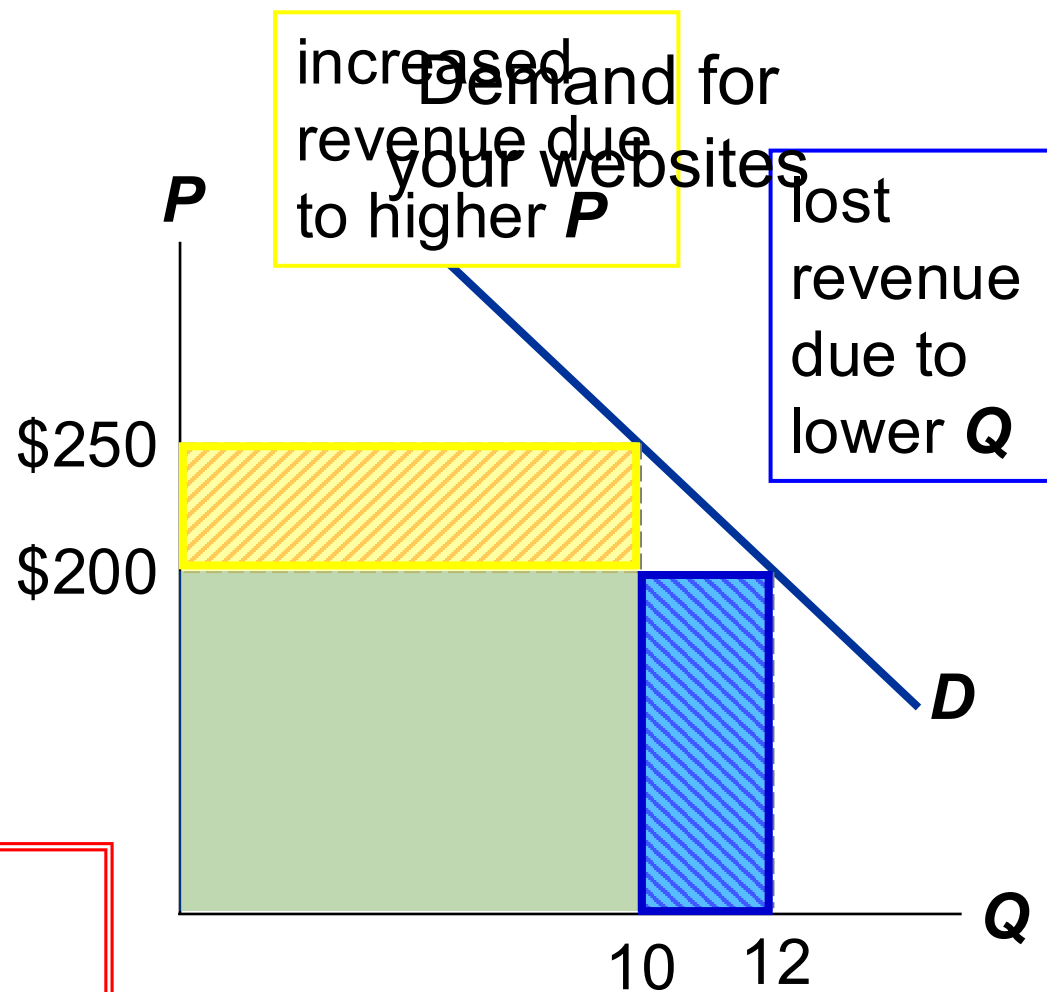
Now, demand is inelastic:

elasticity = 0.82

If $P = \$200$,
 $Q = 12$ and
revenue = \$2400.

If $P = \$250$,
 $Q = 10$ and
revenue = \$2500.

When D is inelastic,
a price increase
causes revenue to rise.



Food for thought

(Elasticity and expenditure/revenue)

A. Pharmacies raise the price of insulin by 10%. Does total expenditure on insulin rise or fall?

$$\text{Expenditure} = P \times Q$$

Since demand is inelastic, Q will fall less than 10%, so **expenditure rises**.

Food for thought

(Elasticity and
expenditure/revenue)

- B.** As a result of a fare war, the price of a luxury cruise falls 20%. Does luxury cruise companies' total revenue rise or fall?

$$\text{Revenue} = P \times Q$$

The fall in P reduces revenue, but Q increases, which increases revenue. Which effect is bigger?

Since demand is elastic, Q will increase more than 20%, so **revenue rises**.

Price Elasticity of Supply

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in } Q^s}{\text{Percentage change in } P}$$

- **Price elasticity of supply** measures how much Q^s responds to a change in P .
- Loosely speaking, it measures sellers' price-sensitivity.
- Again, use the midpoint method to compute the percentage changes.

Price Elasticity of Supply

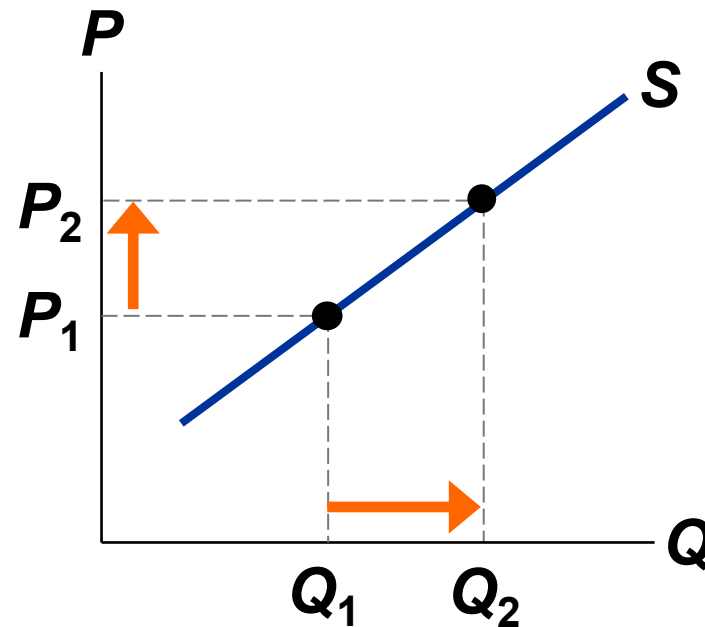
$$\text{Price elasticity of supply} = \frac{\text{Percentage change in } Q^s}{\text{Percentage change in } P}$$

Example:

Price
elasticity
of supply
equals

$$\frac{16\%}{8\%} = 2.0$$

P rises
by 8%



Q rises
by 16%

The Determinants of Supply Elasticity:

- The more easily sellers can change the quantity they produce, the greater the price elasticity of supply.
 - Example: Supply of beachfront property is harder to vary and thus less elastic than supply of new cars.
- For many goods, price elasticity of supply is greater in the long run than in the short run, because firms can build new factories, or new firms may be able to enter the market.

The Variety of Supply Curves

The slope of the supply curve is closely related to price elasticity of supply.

Rule of thumb:

The flatter the curve, the bigger the elasticity.

The steeper the curve, the smaller the elasticity.

Five different classifications of *S* curve...

“Perfectly inelastic” (one extreme)

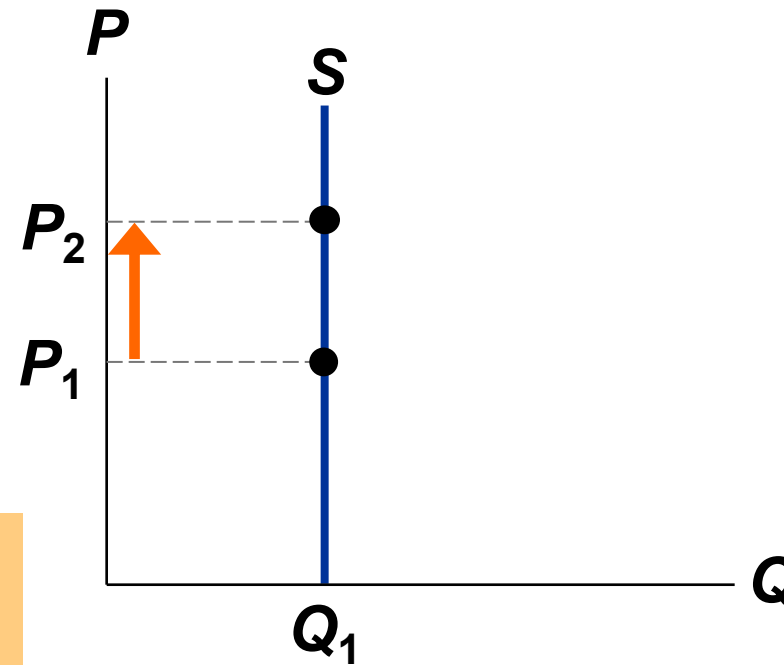
$$\text{Price elasticity of supply} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{0\%}{10\%} = 0$$

S curve:
vertical

Sellers’
price sensitivity:
none

Elasticity:
0

P rises
by 10%



Q changes
by 0%

"Inelastic"

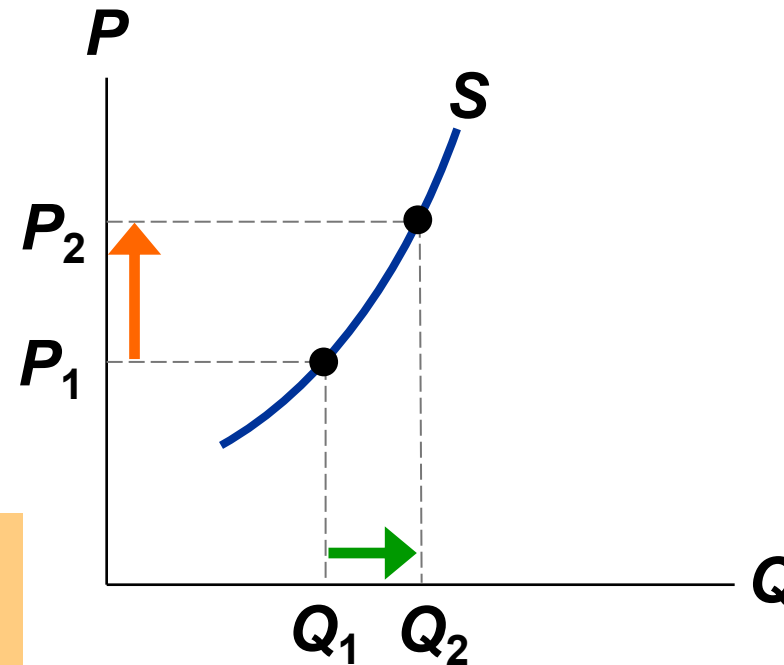
$$\text{Price elasticity of supply} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{< 10\%}{10\%} < 1$$

S curve:
relatively steep

Sellers'
price sensitivity:
relatively low

Elasticity:
< 1

P rises
by 10%



Q rises less
than 10%

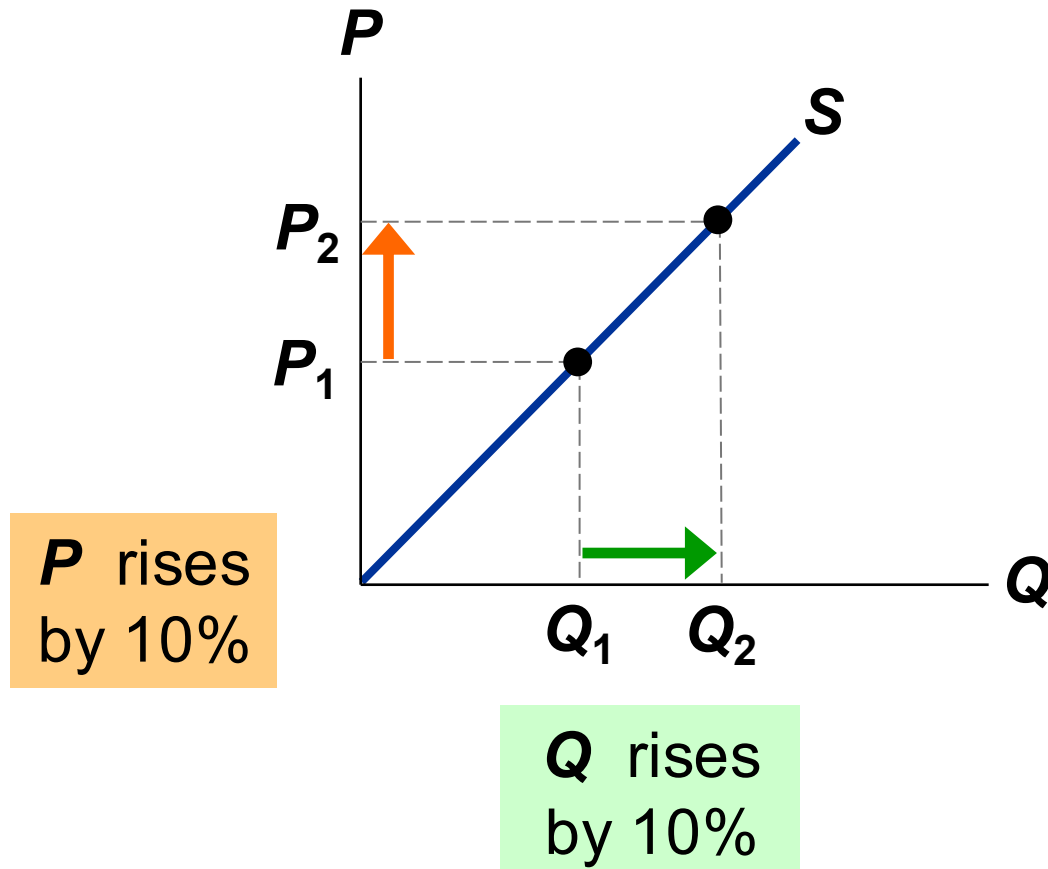
“Unit elastic”

$$\text{Price elasticity of supply} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{10\%}{10\%} = 1$$

S curve:
intermediate slope

Sellers’
price sensitivity:
intermediate

Elasticity:
= 1



"Elastic"

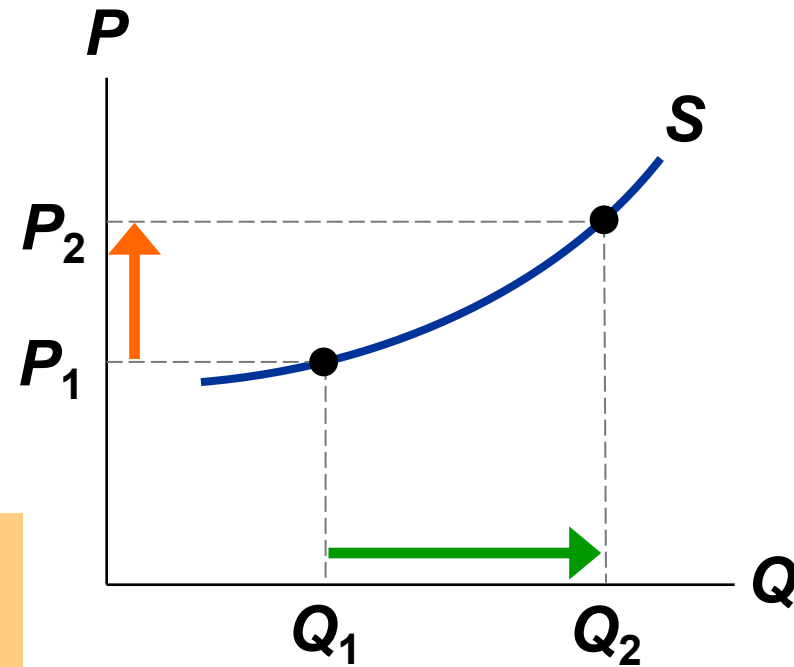
$$\text{Price elasticity of supply} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{> 10\%}{10\%} > 1$$

S curve:
relatively flat

Sellers'
price sensitivity:
relatively high

Elasticity:
> 1

P rises
by 10%



Q rises more
than 10%

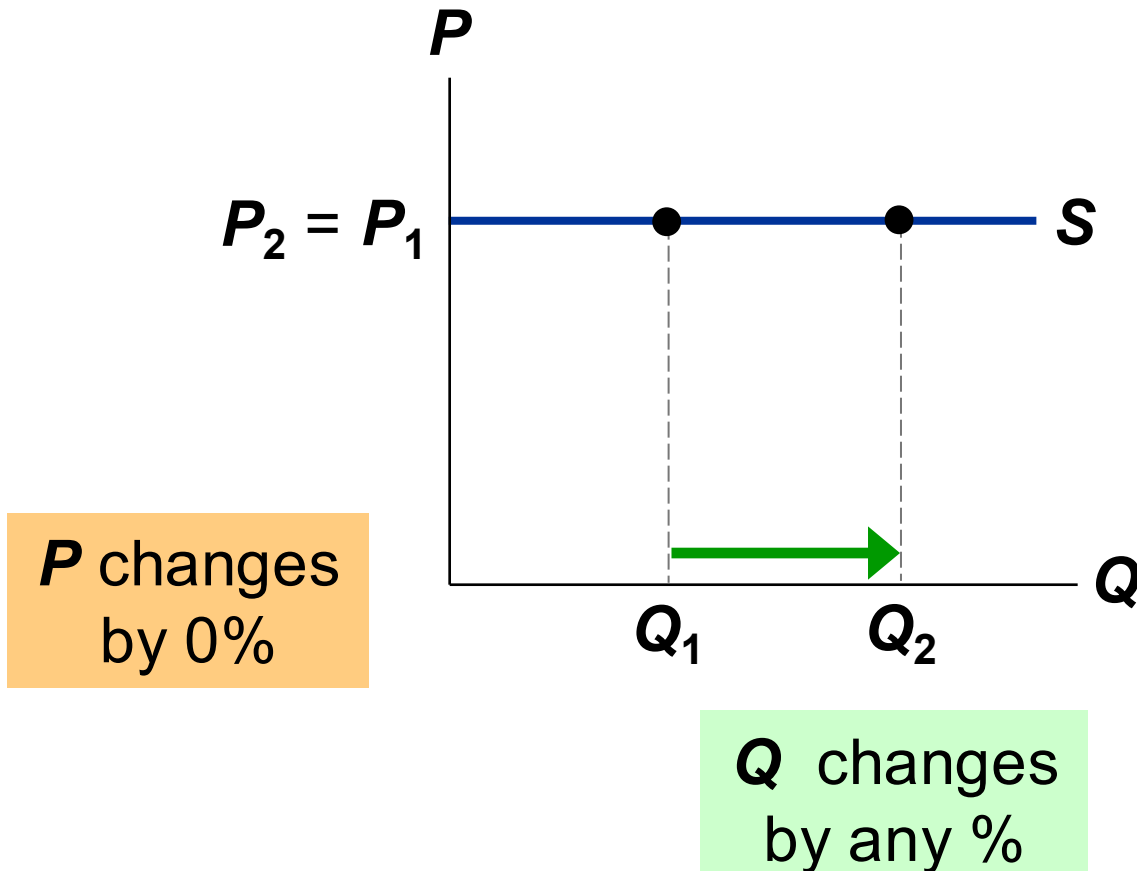
“Perfectly elastic” (the other extreme)

$$\text{Price elasticity of supply} = \frac{\% \text{ change in } Q}{\% \text{ change in } P} = \frac{\text{any } \%}{0\%} = \text{infinity}$$

S curve:
horizontal

Sellers’
price sensitivity:
extreme

Elasticity:
infinity



Food for thought

(Elasticity and changes in equilibrium)

The supply of beachfront property is inelastic.

The supply of new cars is elastic.

Suppose population growth causes demand for both goods to double (at each price, Q^d doubles).

For which product will P change the most?

For which product will Q change the most?

Income Elasticity of Demand

- measures the response of Q^d to a change in consumer income

$$\text{Income elasticity of demand} = \frac{\text{Percent change in } Q^d}{\text{Percent change in income}}$$

- Recall from previous lecture: An increase in income causes an increase in demand for a *normal* good.
- Hence, for normal goods, income elasticity > 0 .
- For *inferior* goods, income elasticity < 0 .

Cross-price Elasticity of Demand

- measures the response of demand for one good to changes in the price of another good

$$\text{Cross-price elast. of demand} = \frac{\% \text{ change in } Q^d \text{ for good 1}}{\% \text{ change in price of good 2}}$$

- For substitutes, cross-price elasticity > 0
(e.g., an increase in price of beef causes an increase in demand for chicken)
- For complements, cross-price elasticity < 0
(e.g., an increase in price of computers causes decrease in demand for software)

Food for thought

Demand and Supply Equations

A market research between Coke and Mojo

A demand schedule for Coke can be represented by the equation

$$QD_c = 200 - 5P_c + 2M + 3P_m ,$$

where QD_c is the Quantity Demanded of Coke, P_c is Price of Coke, M is income and P_m is the Price of Mojo.

The supply schedule for Coke can be represented by the equation

$QSc = 100 + 5P_c$, where QSc is the quantity supplied.

Food for thought

Demand and Supply Equations

- 1.** *Given $M = 80$ and $P_m = 20$, calculate equilibrium Price and Quantity for Coke. [OLD Values]*
- 2.** *When $M = 100$ and $P_m = 20$, calculate equilibrium Price and Quantity for Coke and then calculate YED of Coke*
- 3.** *When $M = 80$ and $P_m = 30$, calculate equilibrium Price and Quantity for Coke and then calculate XED of Coke*

SUMMARY

- **Elasticity** measures the **responsiveness** of Q^d or Q^s to one of its **determinants**.
- Price elasticity of demand equals percentage change in Q^d divided by percentage change in P .
- *When it's less than one, demand is "inelastic."*
When greater than one, demand is "elastic."
- When demand is inelastic, total revenue rises when price rises. When demand is elastic, total revenue falls when price rises.
- Demand is less elastic: in the short run; for necessities; for broadly defined goods; and for goods with few close substitutes.

SUMMARY

- Price elasticity of supply equals percentage change in Q^s divided by percentage change in P .
- *When it's less than one, supply is “inelastic.”*
When greater than one, supply is “elastic.”
- Price elasticity of supply is greater in the long run than in the short run.
- The income elasticity of demand measures how much quantity demanded responds to changes in buyers' incomes.
- The cross-price elasticity of demand measures how much demand for one good responds to changes in the price of another good.